

59

No.

Date

$$P(F|A) = \frac{n(F \cap A)}{n(A)}$$

$$P(F|A) = \frac{P(F \cap A)}{P(A)}$$

1. $n(A) = 60$

$n(B) = 45$

$n(A \cap B) = 25$

$$= \frac{14}{70} = \frac{1}{5} = 20\%$$

Misal $\rightarrow X$ = kejadian minimal mengambil 1. c. $P(F) = P(F \cap A) + P(F \cap B)$

2. $P(X) = \frac{n(A) + n(B) - n(A \cap B)}{100}$

$$= \frac{60 + 45 - 25}{100}$$

$$P(X) = \frac{80}{100} = 80\%$$

$$= \frac{6}{100} + \frac{14}{100} = \frac{20}{100} = 20\%$$

b. $P(Y) = \frac{100 - 20}{100}$

$$= \frac{80}{100} = 80\%$$

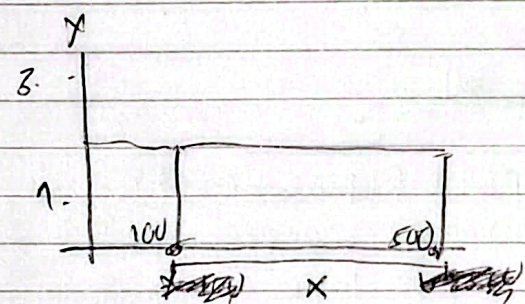
d. $P(F_A) = \frac{n(F_A)}{n(F)}$

$$= \frac{6}{20} = \frac{3}{10} = 30\%$$

c. $P(B \cap A^c) = \frac{n(B) - n(A \cap B)}{100}$

$$= \frac{45 - 25}{100}$$

$$= \frac{20}{100} = 20\%$$



2. $n(F) = 6$

1. $n(A) = 30$

$n(F^c) = 24$

3. $n(B) = 70$

$n(F) = 14$

$n(F^c) = 56$

2. $P(X < 200) = \frac{(200 - 100) \times 1}{400 + 1} = \frac{100}{400} = 25\%$

2. $P(F|A) = \frac{P(F \cap A)}{P(A)}$

$$= \frac{6}{30} = \frac{1}{5} = 20\%$$

b. $P(200 < X < 350) = \frac{350 - 200}{400}$

$$= \frac{150}{400} = \frac{3}{8}$$

c. $P(X = 250) = 0$

b. $P(F|B) = \frac{P(F \cap B)}{P(B)}$

d. karena pada kasus no. 3 ini, Variabel yang digunakan bersifat continuous

yang dimana nilai dari variabel/
sampel tidak terpisah dengan yang
lain, berbeda dengan bilangan diskrit
yang terdiri dari nilai terpisah
yang dapat dihitung.

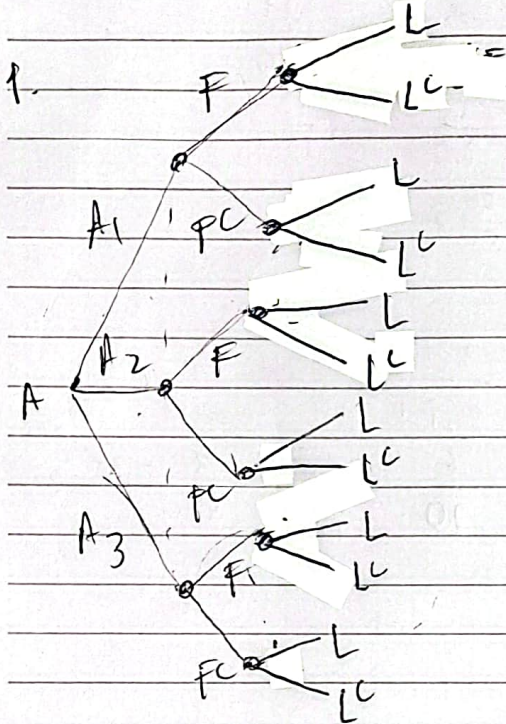
$$1). P(L) = P(A_1 \cap L) + P(A_2 \cap L) + P(A_3 \cap L)$$

$$2). P(F^c) = 1 - P(F) \\ = 1 - 0,091 \\ = 0,909$$

$$P(L) = P(L|F) \cdot P(F) + P(L|F^c) \cdot P(F^c) \\ = 0,95 + 0,041 + 0,04 \times 0,909 \\ = 0,973$$

$P(L|F) \rightarrow$ peluang dalam sistem
memberikan alert jika
koreksi gagal

$P(F|L) \rightarrow$ peluang koreksi gagal
ketika sistem mem-
berikan failure



$$2. P(F) = P(F|A_1) + P(F|A_2) + P(F|A_3) \\ = P(F|A_1) \cdot P(A_1) + P(F|A_2) \cdot P(A_2) + P(F|A_3) \cdot P(A_3) \\ = 0,02 \times 0,15 + 0,05 \times 0,3 + 0,08 \times 0,12 \\ = 0,01 + 0,015 + 0,0096 \\ P(F) = 0,0346$$

$$b. P(F|A_2) = 0,05$$

$$c. P(A_2|F) \times P(F) = P(F|A_2) \times P(A_2)$$

$$P(A_2|F) = \frac{P(F|A_2) \times P(A_2)}{P(F)}$$

$$= \frac{0,05 \times 0,3}{0,0346}$$

$$= 0,3658$$

$$\approx 0,37 //$$

$$e). P(F|L) = \frac{P(F \cap L)}{P(L)}$$

$$= \frac{P(F \cap L)}{P(L)} \times \frac{P(F)}{P(F)}$$

$$= \frac{P(L \cap F)}{P(F)} \times \frac{P(F)}{P(L)}$$

$$= \frac{P(L|F) \times P(F)}{P(L)}$$

$$= \frac{0,95 \times 0,0346}{0,07731} = 0,50 //$$